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Week 12 ICMP

12.2

Ping ke www.ust.hk

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Pinging www.ust.hk [143.89.209.9] with 32 bytes of data:
Reply from 143.89.209.9: bytes=32 time=74ms TTL=42
Reply from 143.89.209.9: bytes=32 time=63ms TTL=42
Reply from 143.89.209.9: bytes=32 time=63ms TTL=42
Reply from 143.89.209.9: bytes=32 time=63ms TTL=42
Reply from 143.89.209.9: bytes=32 time=69ms TTL=42
Request timed out.
Reply from 143.89.209.9: bytes=32 time=63ms TTL=42
Reply from 143.89.209.9: bytes=32 time=63ms TTL=42
Reply from 143.89.209.9: bytes=32 time=61ms TTL=42
Reply from 143.89.209.9: bytes=32 time=61ms TTL=42

Ping statistics for 143.89.209.9:
    Packets: Sent = 10, Received = 9, Lost = 1 (10% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 61ms, Maximum = 74ms, Average = 64ms
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Hasil wireshark

863	38.246901500	192.168.100.90	143.89.209.9	ICMP	74 Echo (ping) request	id=0x0001, seq=45/11520, ttl=128 (reply in 865)
865	38.321524700	143.89.209.9	192.168.100.90	ICMP	74 Echo (ping) reply	id=0x0001, seq=45/11520, ttl=42 (request in 863)
901	39.245987900	192.168.100.90	143.89.209.9	ICMP	74 Echo (ping) request	id=0x0001, seq=46/11776, ttl=128 (reply in 902)
902	39.313355300	143.89.209.9	192.168.100.90	ICMP	74 Echo (ping) reply	id=0x0001, seq=46/11776, ttl=42 (request in 901)
911	40.262520500	192.168.100.90	143.89.209.9	ICMP	74 Echo (ping) request	id=0x0001, seq=47/12032, ttl=128 (reply in 912)
912	40.326278900	143.89.209.9	192.168.100.90	ICMP	74 Echo (ping) reply	id=0x0001, seq=47/12032, ttl=42 (request in 911)
950	41.279019000	192.168.100.90	143.89.209.9	ICMP	74 Echo (ping) request	id=0x0001, seq=48/12288, ttl=128 (reply in 952)
952	41.342351600	143.89.209.9	192.168.100.90	ICMP	74 Echo (ping) reply	id=0x0001, seq=48/12288, ttl=42 (request in 950)
961	42.289040000	192.168.100.90	143.89.209.9	ICMP	74 Echo (ping) request	id=0x0001, seq=49/12544, ttl=128 (reply in 963)
963	42.358100500	143.89.209.9	192.168.100.90	ICMP	74 Echo (ping) reply	id=0x0001, seq=49/12544, ttl=42 (request in 961)
964	43.301454100	192.168.100.90	143.89.209.9	ICMP	74 Echo (ping) request	id=0x0001, seq=50/12800, ttl=128 (no response found!)
1097	47.851045100	192.168.100.90	143.89.209.9	ICMP	74 Echo (ping) request	id=0x0001, seq=51/13056, ttl=128 (reply in 1099)
1099	47.914111100	143.89.209.9	192.168.100.90	ICMP	74 Echo (ping) reply	id=0x0001, seq=51/13056, ttl=42 (request in 1097)
1120	48.862152500	192.168.100.90	143.89.209.9	ICMP	74 Echo (ping) request	id=0x0001, seq=52/13312, ttl=128 (reply in 1126)
1126	48.924933300	143.89.209.9	192.168.100.90	ICMP	74 Echo (ping) reply	id=0x0001, seq=52/13312, ttl=42 (request in 1120)
1152	49.871632700	192.168.100.90	143.89.209.9	ICMP	74 Echo (ping) request	id=0x0001, seq=53/13568, ttl=128 (reply in 1153)
1153	49.933352300	143.89.209.9	192.168.100.90	ICMP	74 Echo (ping) reply	id=0x0001, seq=53/13568, ttl=42 (request in 1152)
1158	50.880791000	192.168.100.90	143.89.209.9	ICMP	74 Echo (ping) request	id=0x0001, seq=54/13824, ttl=128 (reply in 1159)
1159	50.942448200	143.89.209.9	192.168.100.90	ICMP	74 Echo (ping) reply	id=0x0001, seq=54/13824, ttl=42 (request in 1158)
1310	54.402529300	182.8.99.30	192.168.100.90	ICMP	207 Destination unreachable (Port unreachable)	
1317	54.407275800	182.8.99.30	192.168.100.90	ICMP	207 Destination unreachable (Port unreachable)	
1318	54.407275800	182.8.99.30	192.168.100.90	ICMP	207 Destination unreachable (Port unreachable)	

Penjelasan :

1. Host 192.168.100.90 berhasil melakukan komunikasi ICMP dengan 143.89.209.9.
2. Paket ICMP Echo Request dan Echo Reply berhasil dipertukarkan.
3. Nilai TTL menunjukkan paket melewati banyak router sebelum mencapai tujuan.
4. Terdapat satu paket yang mengalami timeout/no response pada sequence 50, kemungkinan akibat packet loss atau delay jaringan.

12.3

Tracert

```
Command Prompt
7 185 ms 174 ms 174 ms 180.240.196.1
8 184 ms 184 ms 185 ms renater.par.franceix.net [37.49.236.19]
9 185 ms 185 ms 185 ms hu0-4-0-1-ren-nr-orsay-rtr-091.noc.renater.fr [193.51.180.43]
10 196 ms 197 ms 196 ms inria-rocquencourt-vl1631-tel-4-inria-rtr-021.noc.renater.fr [193.51.184.177]
11 185 ms 185 ms 185 ms unit240-reth1-vfw-ext-dcl.inria.fr [192.93.122.19]
12 197 ms 197 ms 197 ms prod-inriafr-cms.inria.fr [128.93.162.83]

Trace complete.

C:\Users\USER>tracert www.inria.fr

Tracing route to inria.fr [128.93.162.83]
over a maximum of 30 hops:

 1  12 ms  14 ms  33 ms  192.168.100.1
 2  12 ms  4 ms  4 ms  10.114.0.1
 3  4 ms  4 ms  4 ms  180.250.252.69
 4  31 ms  22 ms  23 ms  180.240.190.77
 5  29 ms  23 ms  22 ms  180.240.190.77
 6  174 ms  174 ms  176 ms  180.240.192.233
 7  181 ms  173 ms  174 ms  180.240.196.1
 8  193 ms  185 ms  185 ms  renater.par.franceix.net [37.49.236.19]
 9  186 ms  185 ms  186 ms  hu0-4-0-1-ren-nr-orsay-rtr-091.noc.renater.fr [193.51.180.43]
10  197 ms  197 ms  200 ms  inria-rocquencourt-vl1631-tel-4-inria-rtr-021.noc.renater.fr [193.51.184.177]
11  185 ms  185 ms  185 ms  unit240-reth1-vfw-ext-dcl.inria.fr [192.93.122.19]
12  197 ms  196 ms  196 ms  prod-inriafr-cms.inria.fr [128.93.162.83]

Trace complete.
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Hasil wireshark

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Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No. Time Source Destination Protocol Length Info
21 2.452314580 182.8.99.30 192.168.100.90 ICMP 207 Destination unreachable (Port unreachable)
22 2.452415880 182.8.99.30 192.168.100.90 ICMP 207 Destination unreachable (Port unreachable)
23 2.452415880 182.8.99.30 192.168.100.90 ICMP 207 Destination unreachable (Port unreachable)
30 2.733582600 192.168.100.90 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=91/23296, ttl=1 (no response found)
32 2.726281200 192.168.100.1 192.168.100.90 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
33 2.727142600 192.168.100.90 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=92/23552, ttl=1 (no response found)
34 2.727142600 192.168.100.1 192.168.100.90 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
35 2.742673400 192.168.100.90 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=93/23808, ttl=1 (no response found)
37 2.776315200 192.168.100.1 192.168.100.90 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
197 8.715343600 192.168.100.90 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=94/24064, ttl=2 (no response found)
200 8.728046000 10.114.0.1 192.168.100.90 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
201 8.728781200 192.168.100.90 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=95/24320, ttl=2 (no response found)
202 8.732658800 10.114.0.1 192.168.100.90 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
203 8.733414600 192.168.100.90 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=96/24576, ttl=2 (no response found)
204 8.733865200 10.114.0.1 192.168.100.90 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)

Frame 32: Packet, 134 bytes on wire (1072 bits), 134 bytes captured (1072 bits) on interface \Device\NPF...
Ethernet II, Src: HuaweiTechno_65:14:cc (d4:d5:1b:65:14:cc), Dst: WNC_3a:3f:ea (40:1a:58:3a:3f:ea)
Internet Protocol Version 4, Src: 192.168.100.1, Dst: 192.168.100.90
Internet Control Message Protocol
Type: Time-to-live exceeded (11)
Code: 0 (Time to live exceeded in transit)
Checksum: 0xf4ff [correct]
[Checksum Status: Good]
Unused: 00000000
Internet Protocol Version 4, Src: 192.168.100.90, Dst: 128.93.162.83
Internet Control Message Protocol
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